Functions, Rational Expressions & Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • [FRQ] • Target: Simplifying Rational Expressions & Domain Restrictions**

Factor the numerator and denominator completely, state all domain restrictions, and simplify $\frac{2x^2-5x-12}{x^2-16}$.

Factor: $\frac{(2x+3)(x-4)}{(x-4)(x+4)}$. Domain restrictions: $x \neq 4$ and $x \neq -4$. Canceling $x-4$ yields $\frac{2x+3}{x+4}$ for $x \neq \pm 4$.

2 **[Foundational] • [FRQ] • Target: Adding & Subtracting Rational Expressions**

Combine over a common denominator and simplify into a single reduced fraction: $\frac{3}{x-2} - \frac{2x-1}{x^2-4}$.

Since $x^2-4=(x-2)(x+2)$, rewrite as $\frac{3(x+2)-(2x-1)}{(x-2)(x+2)} = \frac{3x+6-2x+1}{x^2-4} = \frac{x+7}{x^2-4}$ $(x \neq \pm 2)$.

3 **[Foundational] • [FRQ] • Target: Solving Rational Equations**

Solve the rational equation $\frac{4}{x-1} + \frac{2}{x+3} = 1$ for all valid real solutions.

Multiply by $(x-1)(x+3)$: $4(x+3)+2(x-1)=(x-1)(x+3) \Rightarrow 6x+10=x^2+2x-3 \Rightarrow x^2-4x-13=0 \Rightarrow x=2\pm\sqrt{17}$ (neither equals 1 or -3).

4 **[Foundational] • [FRQ] • Target: Complex Number Arithmetic**

Perform each complex number operation and write the result in standard form $a+bi$:

a $(5-3i)(2+4i)$ $10+20i-6i-12i^2 = 10+14i+12 = 22+14i$

b $(3-2i)^2-(4+5i)$ $(9-12i+4i^2)-4-5i = (5-12i)-4-5i = 1-17i$

Use $i^2=-1$ and combine real and imaginary parts.

5 **[Foundational] • [FRQ] • Target: Complex Conjugates & Division**

Divide the complex numbers $\frac{14-5i}{3+2i}$ by multiplying numerator and denominator by the complex conjugate, and express the result in the form $a+bi$.

$\frac{(14-5i)(3-2i)}{3^2+2^2} = \frac{42-28i-15i+10i^2}{13} = \frac{32-43i}{13} = \frac{32}{13} - \frac{43}{13}i$.

6 **[Foundational] • [FRQ] • Target: Powers of the Imaginary Unit i**

Simplify $i^{45}+i^{102}-i^{79}$ using the periodicity of powers of i ($i^4=1$).

Divide each exponent by 4: $45=4(11)+1 \Rightarrow i^{45}=i$; $102=4(25)+2 \Rightarrow i^{102}=i^2=-1$; $79=4(19)+3 \Rightarrow i^{79}=i^3=-i$. Thus $i+(-1)-(-i)=-1+2i$.

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7 [Foundational] • [FRQ] • Target: Asymptotes & Intercepts of Rational Functions

State the vertical asymptote, horizontal asymptote, x-intercept, and y-intercept of the rational function $f(x) = \frac{2x-6}{x-1}$ graphed below.

Vertical asymptote: $x = 1$; horizontal asymptote: $y = \frac{2}{1} = 2$; x-intercept: $2x - 6 = 0 \Rightarrow (3, 0)$; y-intercept: $f(0) = 6 \Rightarrow (0, 6)$.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Extraneous Solutions in Rational Equations

Solve the rational equation $\frac{x}{x-3} - \frac{3}{x-3} = 2$ and explain why it has no solution.

Multiplying both sides by $x - 3$ (for $x \neq 3$) gives $x - 3 = 2(x - 3) \Rightarrow x = 3$. However, $x = 3$ makes the denominator $x - 3 = 0$, so $x = 3$ is an extraneous solution and the equation has no solution.

9 [Intermediate] • [FRQ] • Target: Partial Fraction Decomposition

Find the constants A and B in the partial-fraction identity $\frac{5x+1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$.

Multiply by $(x - 1)(x + 2)$: $5x + 1 = A(x + 2) + B(x - 1)$. Setting $x = 1$ gives $6 = 3A \Rightarrow A = 2$. Setting $x = -2$ gives $-9 = -3B \Rightarrow B = 3$.

10 [Intermediate] • [FRQ] • Target: Simplifying Complex Fractions

Simplify the complex fraction $\frac{\frac{1}{x} - \frac{1}{3}}{x-3}$ for $x \neq 0, 3$, and evaluate the simplified expression at $x = 3$.

Combine the numerator: $\frac{\frac{3-x}{3x}}{x-3} = \frac{-(x-3)}{3x(x-3)} = -\frac{1}{3x}$. Evaluating at $x = 3$ gives $-\frac{1}{9}$.

11 [Intermediate] • [FRQ] • Target: Complex Roots of Quadratic Equations

Find the complex conjugate solutions to $x^2 - 6x + 25 = 0$ in the form $a \pm bi$, and verify that their sum is 6 and their product is 25.

Complete the square: $(x - 3)^2 = -16 \Rightarrow x - 3 = \pm 4i \Rightarrow x = 3 \pm 4i$. Sum $= (3 + 4i) + (3 - 4i) = 6$; Product $= 3^2 + 4^2 = 9 + 16 = 25$.

12 [Intermediate] • [FRQ] • Target: Quotient-Remainder Form of Rational Functions

Use polynomial long division to rewrite $\frac{2x^2+5x-1}{x+3}$ in the form $ax + b + \frac{r}{x+3}$.

Divide $2x^2 + 5x - 1$ by $x + 3$: $2x^2 + 5x - 1 = (2x - 1)(x + 3) + 2$. Therefore $\frac{2x^2+5x-1}{x+3} = 2x - 1 + \frac{2}{x+3}$.

13 [Intermediate] • [FRQ] • Target: Harmonic / Parallel Resistance Formulas

Two resistors R_1 and R_2 connected in parallel have equivalent resistance R_{eq} given by $\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$. Solve for R_2 in terms of R_{eq} and R_1, and compute R_2 when $R_{\text{eq}} = 12\,\Omega$ and $R_1 = 20\,\Omega$.

$\frac{1}{R_2} = \frac{1}{R_{\text{eq}}} - \frac{1}{R_1} = \frac{R_1 - R_{\text{eq}}}{R_{\text{eq}} R_1} \Rightarrow R_2 = \frac{R_{\text{eq}} R_1}{R_1 - R_{\text{eq}}}$. For $R_{\text{eq}} = 12, R_1 = 20$: $R_2 = \frac{12(20)}{20-12} = \frac{240}{8} = 30\,\Omega$.

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14 [Intermediate] • [FRQ] • Target: Solving Rational Inequalities

Solve the rational inequality $\frac{x-4}{x+2} \leq 0$ and express the solution set in interval notation.

The critical values are the zero $x = 4$ (included) and the vertical asymptote $x = -2$ (strictly excluded, since division by zero is undefined). Sign analysis shows $\frac{x-4}{x+2} \leq 0$ on $(-2, 4]$.

Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • [FRQ] • Target: Complex Conjugate Root Theorem • Cross-Lesson: Topic 4.01 (Cubic Polynomials)

A polynomial $P(x)$ of degree 3 with real coefficients has zeros $x = 2$ and $x = 1 + 3i$, and satisfies $P(0) = -40$. Find the factored and expanded forms of $P(x)$.

By the Complex Conjugate Root Theorem, $1 - 3i$ is also a zero. The quadratic factor for $1 \pm 3i$ is $(x-1)^2 + 9 = x^2 - 2x + 10$. Thus $P(x) = a(x-2)(x^2 - 2x + 10)$. Since $P(0) = -20a = -40 \Rightarrow a = 2$, we get $P(x) = 2(x-2)(x^2 - 2x + 10) = 2x^3 - 8x^2 + 28x - 40$.

16 [Challenging] • [FRQ] • Target: Removable Discontinuities vs. Vertical Asymptotes • Cross-Lesson: Topic 3.01 (Quadratic Factoring)

Find the domain, removable discontinuity (hole) coordinates, and vertical asymptote of $g(x) = \frac{x^2 - 9}{x^2 - x - 6}$.

Factor numerator and denominator: $g(x) = \frac{(x-3)(x+3)}{(x-3)(x+2)}$. The domain is $x \neq -2, 3$. The factor $x - 3$ cancels, giving $g(x) = \frac{x+3}{x+2}$ for $x \neq 3$: thus there is a removable discontinuity (hole) at $\left(3, \frac{6}{5}\right)$ and a vertical asymptote at $x = -2$.

17 [Challenging] • [FRQ] • Target: Upstream-Downstream Rational Modeling • Cross-Lesson: Topic 5.01 (Rates) & Topic 3.01 (Quadratics)

A kayaker paddles $12\,\text{km}$ upstream and $12\,\text{km}$ back downstream in a river whose current flows at $3\,\text{km/h}$. If the total round-trip time is 3 hours, set up a rational equation to find the kayaker's speed v in still water.

Time equation: $\frac{12}{v-3} + \frac{12}{v+3} = 3$. Multiply by $(v-3)(v+3) = v^2 - 9$:

$12(v+3) + 12(v-3) = 3(v^2 - 9) \Rightarrow 24v = 3v^2 - 27 \Rightarrow v^2 - 8v - 9 = 0 \Rightarrow (v-9)(v+1) = 0$. Since $v > 3$, $v = 9\,\text{km/h}$.

18 [Challenging] • [FRQ] • Target: Inverting Linear Fractional Transformations • Cross-Lesson: Topic 1.01 (Isolating Variables)

For a one-to-one rational function $f(x) = \frac{3x+2}{x-4}$ $(x \neq 4)$, find the formula for its inverse function $f^{-1}(x)$ and state the domain and range of f.

Set $y = \frac{3x+2}{x-4} \Rightarrow y(x-4) = 3x + 2 \Rightarrow x(y-3) = 4y + 2 \Rightarrow x = \frac{4y+2}{y-3}$. Thus $f^{-1}(x) = \frac{4x+2}{x-3}$. The domain of f is $x \neq 4$ and the range of f (domain of f^{-1}) is $y \neq 3$.

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19 [Challenging] • [FRQ] • Target: Modulus Algebra in the Complex Plane • Cross-Lesson: Topic 7.01 (Pythagorean Theorem)

In the complex plane, find the modulus $|z|$ of $z = \frac{(3+4i)(1-i)}{2-2i}$ without expanding the numerator.

Using the multiplicative property of moduli $|z_1 z_2 / z_3| = |z_1||z_2|/|z_3|$: $|3+4i| = 5$, $|1-i| = \sqrt{2}$, and $|2-2i| = 2\sqrt{2}$. Therefore $|z| = \frac{5\sqrt{2}}{2\sqrt{2}} = \frac{5}{2} = 2.5$.

20 [Challenging] • [FRQ] • Target: Rational-Reciprocal Identities & Optimization • Cross-Lesson: Topic 3.01 (Completing the Square)

If $x > 0$ and $x + \frac{4}{x} = 6$, find the exact value of $x^2 + \frac{16}{x^2}$ and the minimum possible value of $t + \frac{4}{t}$ for all $t > 0$.

Square $x + \frac{4}{x} = 6$: $x^2 + 8 + \frac{16}{x^2} = 36 \Rightarrow x^2 + \frac{16}{x^2} = 28$. For any $t > 0$, $t + \frac{4}{t} = \left(\sqrt{t} - \frac{2}{\sqrt{t}}\right)^2 + 4 \geq 4$, with minimum value 4 achieved at $t = 2$.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Simplifying Rational Expressions

Given the algebraic relationship in which for $x \neq 3$, which expression is equivalent to $\frac{x^2-9}{x-3}$?

a. $x - 3$

b. $x + 3$ ✓

c. $x^2 + 3$

d. $x - 9$

Factor the difference of squares: $\frac{(x-3)(x+3)}{x-3} = x + 3$.

22 [Foundational] • [MCQ] • Target: Adding Complex Numbers

If $i = \sqrt{-1}$, what is the sum $(7 - 4i) + (-2 + 9i)$?

a. $5 + 5i$ ✓

b. $5 - 13i$

c. $9 + 5i$

d. $9 - 5i$

Combine real parts and imaginary parts: $(7 - 2) + (-4 + 9)i = 5 + 5i$.

23 [Foundational] • [MCQ] • Target: Vertical Asymptotes

What is the vertical asymptote of the graph of $f(x) = \frac{5}{2x+8}$?

a. $x = -4$ ✓

b. $x = 0$

c. $x = 4$

d. $x = 5$

Set the denominator equal to zero: $2x + 8 = 0 \Rightarrow x = -4$.

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24 [Foundational] • [MCQ] • Target: Proportions & Rational Equations

Within the real coordinate domain, to satisfy the synthesis condition, if $\frac{6}{x-2} = \frac{3}{4}$, what is the value of x?

a. 6

b. 8

c. 10 ✓

d. 12

Cross-multiply: $3(x-2) = 24 \Rightarrow x - 2 = 8 \Rightarrow x = 10$.

25 [Foundational] • [MCQ] • Target: Product of Complex Conjugates

What is the product $(4+3i)(4-3i)$?

a. 7

b. $16 - 9i$

c. 25 ✓

d. $7 + 24i$

Multiplying complex conjugates gives $a^2 + b^2 = 4^2 + 3^2 = 16 + 9 = 25$.

26 [Foundational] • [MCQ] • Target: Adding Fractions with Variable Denominators

Combine the rational terms $\frac{2}{x} + \frac{5}{3x}$ $(x \neq 0)$ into a single simplified fraction.

a. $\frac{7}{4x}$

b. $\frac{7}{3x}$

c. $\frac{11}{3x}$ ✓

d. $\frac{10}{3x^2}$

Use the common denominator $3x$: $\frac{6}{3x} + \frac{5}{3x} = \frac{11}{3x}$.

27 [Foundational] • [MCQ] • Target: Powers of i

What is the value of i^{26}?

a. -1 ✓

b. 1

c. $-i$

d. i

Since $26 = 4(6) + 2$, we have $i^{26} = (i^4)^6 \cdot i^2 = 1 \cdot (-1) = -1$.

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28 [Foundational] • [MCQ] • Target: Domain of Rational Functions

For what value of x is the function $h(x) = \frac{x+5}{x^2-6x+9}$ undefined?

a. -5 **b.** -3

c. 0 **d.** 3 ✓

The denominator factors as $(x-3)^2 = 0$, so $h(x)$ is undefined at $x = 3$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

29 [Intermediate] • [MCQ] • Target: Rewriting Improper Rational Expressions

Rewrite the improper rational function $\frac{3x+7}{x+2}$ $(x \neq -2)$ in quotient-plus-remainder form.

a. $3 + \frac{1}{x+2}$ ✓ **b.** $3 + \frac{7}{x+2}$

c. $3 + \frac{5}{x+2}$ **d.** $3 - \frac{1}{x+2}$

Rewrite the numerator as $3(x+2)+1$: $\frac{3(x+2)+1}{x+2} = 3 + \frac{1}{x+2}$.

30 [Intermediate] • [MCQ] • Target: Complex Division & Real/Imaginary Parts

Analyze the configuration in which when $\frac{10}{1-2i}$ is written in the form $a+bi$, what is the value of $a+b$?

a. 2 **b.** 4

c. 6 ✓ **d.** 8

Multiply by $\frac{1+2i}{1+2i}$: $\frac{10(1+2i)}{1^2+2^2} = \frac{10(1+2i)}{5} = 2+4i$. Thus $a+b = 2+4 = 6$.

31 [Intermediate] • [MCQ] • Target: Filtering Extraneous Roots

What are all real solutions to $\frac{x^2}{x-4} = \frac{16}{x-4}$?

a. $x = 4$ only **b.** $x = -4$ **only** ✓

c. $x = 4$ and $x = -4$ **d.** No real solution

For $x \neq 4$, $x^2 = 16 \Rightarrow x = \pm 4$. Since $x = 4$ makes the denominator zero (extraneous), the only valid solution is $x = -4$.

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32 [Intermediate] • [MCQ] • Target: Horizontal Asymptotes of Equal-Degree Rationals

What is the horizontal asymptote of $f(x) = \frac{6x^2 - 5x + 1}{2x^2 + 9}$?

a. $y = 0$

b. $y = \frac{1}{3}$

c. $y = 3$ ✓

d. $y = 6$

Since numerator and denominator have equal degree (2), the horizontal asymptote is the ratio of leading coefficients: $y = \frac{6}{2} = 3$.

33 [Intermediate] • [MCQ] • Target: Partial Fractions Cover-Up Method

If $\frac{4x - 2}{x^2 - 1} = \frac{A}{x - 1} + \frac{B}{x + 1}$ for all $x \neq \pm 1$, what is the value of A?

a. 1 ✓

b. 2

c. 3

d. 4

Multiply by $x^2 - 1$: $4x - 2 = A(x + 1) + B(x - 1)$. Substitute $x = 1$: $2 = 2A \Rightarrow A = 1$ (and at $x = -1$, $-6 = -2B \Rightarrow B = 3$).

34 [Intermediate] • [MCQ] • Target: Simplifying Compound Fractions

Simplify the compound reciprocal fraction $\frac{\frac{1}{a} + \frac{1}{b}}{\frac{1}{ab}}$ for positive real numbers $a, b > 0$.

a. $\frac{1}{a + b}$

b. $a + b$ ✓

c. ab

d. $\frac{a + b}{a^2 b^2}$

Multiply numerator and denominator by ab: $(b + a)/1 = a + b$.

35 [Intermediate] • [MCQ] • Target: Equating Real & Imaginary Parts

If $(x + yi)(2 - i) = 8 + i$, where x and y are real numbers, what is the value of $x + y$?

a. 3

b. 4

c. 5 ✓

d. 6

Divide $8 + i$ by $2 - i$: $\frac{(8 + i)(2 + i)}{2^2 + 1^2} = \frac{16 + 10i - 1}{5} = \frac{15 + 10i}{5} = 3 + 2i$. Thus $x = 3$ and $y = 2$, giving $x + y = 5$.

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36 [Intermediate] • [MCQ] • Target: Rational Expressions for Combined Work Rates

Printer A can print a batch of reports in x hours, and Printer B can print the same batch in $x + 3$ hours. Which expression represents the fraction of the batch they can print together in 1 hour?

a. $\frac{2}{2x+3}$

b. $\frac{2x+3}{x(x+3)}$ ✓

c. $\frac{x(x+3)}{2x+3}$

d. $\frac{1}{x(x+3)}$

Their combined hourly rate is $\frac{1}{x} + \frac{1}{x+3} = \frac{(x+3)+x}{x(x+3)} = \frac{2x+3}{x(x+3)}$.

37 [Intermediate] • [MCQ] • Target: Reconstructing Quadratics from Complex Conjugate Roots

Which of the following quadratic equations has roots $2 + i$ and $2 - i$?

a. $x^2 - 4x + 3 = 0$

b. $x^2 - 4x + 5 = 0$ ✓

c. $x^2 + 4x + 5 = 0$

d. $x^2 - 2x + 5 = 0$

Sum of roots $= (2 + i) + (2 - i) = 4$; product of roots $= 2^2 + 1^2 = 5$. Thus the monic quadratic is $x^2 - 4x + 5 = 0$.

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Periodic Series of Powers of i • Cross-Lesson: Topic 4.01 (Exponents & Periodicity)

What is the value of the sum $i^1 + i^2 + i^3 + \ldots + i^{99}$?

a. -1 ✓

b. 0

c. i

d. $-i$

Every block of 4 consecutive powers of i sums to $i - 1 - i + 1 = 0$. Since $99 = 4(24) + 3$, the first 96 terms sum to 0, leaving $i^{97} + i^{98} + i^{99} = i + (-1) + (-i) = -1$.

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39 [Challenging] • [MCQ] • Target: Hyperbola-Line Intersection System • Cross-Lesson: Topic 2.01 (Systems) & Topic 3.01 (Quadratics)

In the xy-plane, at how many points does the rational curve $y = \frac{6}{x}$ intersect the line $x + y = 5$?

a. Zero

b. One

c. **Two ✓**

d. Three

Substitute $y = 5 - x$: $\frac{6}{x} = 5 - x \Rightarrow x^2 - 5x + 6 = 0 \Rightarrow (x - 2)(x - 3) = 0$. This gives two intersection points: $(2, 3)$ and $(3, 2)$.

40 [Challenging] • [MCQ] • Target: Rational Dilution Modeling & Asymptotic Behavior • Cross-Lesson: Topic 5.01 (Mixture & Concentration)

A chemistry student adds x liters of pure water to 20 liters of a 40% acid solution. Which rational function gives the new acid concentration $C(x)$ as a decimal, and what is $\lim_{x \to \infty} C(x)$?

a. $C(x) = \frac{8}{20 + x}$**, horizontal asymptote** $C = 0$ ✓

b. $C(x) = \frac{8 + x}{20 + x}$, horizontal asymptote $C = 1$

c. $C(x) = \frac{0.40}{20 + x}$, horizontal asymptote $C = 0$

d. $C(x) = \frac{20}{8 + x}$, horizontal asymptote $C = 0$

Pure acid remains constant at $0.40(20) = 8$ liters, while total volume becomes $20 + x$ liters, so $C(x) = \frac{8}{20 + x} \to 0$ as $x \to \infty$.

41 [Challenging] • [MCQ] • Target: Integer Divisibility via Rational Division • Cross-Lesson: Topic 4.01 (Polynomial Division)

For how many integer values of n is $\frac{2n + 15}{n + 2}$ also an integer?

a. 2

b. 4 ✓

c. 6

d. 8

Divide: $\frac{2n + 15}{n + 2} = \frac{2(n + 2) + 11}{n + 2} = 2 + \frac{11}{n + 2}$. This is an integer if and only if $n + 2$ divides the prime 11, i.e. $n + 2 \in \{\pm 1, \pm 11\}$ (4 integer values of n: $-13, -3, -1, 9$).

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42 [Challenging] • [MCQ] • Target: 90-Degree Rotation by i in the Complex Plane • Cross-Lesson: Topic 7.01 (45-45-90 Right Triangles)

If $z = 3 + 4i$, what is the distance in the complex plane between z and iz?

a. 5

b. $5\sqrt{2}$ ✓

c. 7

d. 10

Multiplying by i rotates z by 90° around the origin without changing its length $|z| = 5$ (specifically, $iz = -4 + 3i$). By the $45^\circ-45^\circ-90^\circ$ right triangle with legs 5 and 5, the distance $|z - iz|$ is $5\sqrt{2}$.

43 [Challenging] • [MCQ] • Target: Non-Zero Right-Hand-Side Rational Inequalities • Cross-Lesson: Topic 2.01 (Inequalities)

What is the solution set to the rational inequality $\frac{x+5}{x-2} \geq 2$?

a. $(2, 9]$ ✓

b. $[9, \infty)$

c. $(-\infty, 2) \cup [9, \infty)$

d. $[-5, 2)$

Subtract 2: $\frac{x+5-2(x-2)}{x-2} = \frac{9-x}{x-2} \geq 0 \Leftrightarrow \frac{x-9}{x-2} \leq 0$. Since $x = 2$ is excluded and $x = 9$ is included, the solution set is $(2, 9]$.

44 [Challenging] • [MCQ] • Target: Inverse Pythagorean Theorem for Right-Triangle Altitude • Cross-Lesson: Topic 7.01 (Right-Triangle Altitude)

A right triangle has legs of lengths $\frac{1}{a}$ and $\frac{1}{b}$, where $a, b > 0$. Which rational expression gives the length of the altitude drawn to the hypotenuse?

a. $\frac{1}{a+b}$

b. $\frac{1}{\sqrt{a^2+b^2}}$ ✓

c. $\frac{ab}{\sqrt{a^2+b^2}}$

d. $\frac{\sqrt{a^2+b^2}}{ab}$

The inverse-Pythagorean altitude formula states $\frac{1}{h^2} = \frac{1}{leg_1^2} + \frac{1}{leg_2^2} = a^2 + b^2 \Rightarrow h = \frac{1}{\sqrt{a^2+b^2}}$.

45 [Challenging] • [MCQ] • Target: Iterated Composition of Rational Functions • Cross-Lesson: Topic 4.01 (Function Composition)

If $f(x) = \frac{x-1}{x+1}$, what is $f(f(x))$ for all $x \neq -1, 0$?

a. x

b. $-\frac{1}{x}$ ✓

c. $\frac{1}{x}$

d. $-x$

$f(f(x)) = \frac{\frac{x-1}{x+1} - 1}{\frac{x-1}{x+1} + 1} = \frac{(x-1)-(x+1)}{(x-1)+(x+1)} = \frac{-2}{2x} = -\frac{1}{x}$.

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